
DAWN-SRW: State-Conditioned Programs for Learned State Transitions

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Abstract

Learned state-transition computation is often organizationally implicit. We ask whether it can instead be organized around explicit learned operations and state-conditioned applicability, with their controlled contributions composed into transition-producing computation and exposed for measurement and causal intervention. We study this question constructively in autoregressive language models. We use $(\mathcal{O}, C, \Gamma)$ as minimal organizational coordinates for one family of explicit learned transition organizations. Within these coordinates, a *State-Conditioned Program* is one program-oriented formulation in which controlled operation contributions are constituents of actual forward execution. It is not a claim of symbolic program text, formal operational semantics, or an intrinsic or privileged neural decomposition. DAWN-SRW provides one constructive realization: continuous geometric applicability over persistent rank-1 read-write (RW) operators controls which full-dimensional RW responses contribute and by how much, while normalized additive composition forms the transition-producing output. The evaluated model retains full candidate scope and dense RW execution. At 393.8M parameters and 39.999B tokens, DAWN-SRW supports viability relative to the matched dense control: C4 cross-entropy is 2.9402 versus 2.9359 and the six-task zero-shot mean is 0.4629 versus 0.4597. State-conditioned exact applicability support provides evidence of conditional organization. A discovery-frozen 4,096-site Q/K-centered operator circuit passes suppression, four matched-control families, and exact restoration on validation and held-out indirect-object-identification (IOI) inputs, establishing causal accessibility; fail-closed results delimit unsuccessful localization. They do not establish human-readable operator semantics, benchmark superiority, hardware efficiency, or an intrinsic or unique decomposition.

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1. Introduction

Learned systems compute through state transitions, but their transition-producing organization is often implicit. State dependence and conditional structure are established; the broader problem is whether learned computation can be architecturally organized around explicit learned operations and state-conditioned applicability, with controlled contributions composed into transition-producing computation. Autoregressive language models provide a concrete test setting in which high-dimensional representations undergo repeated transitions. If X_ℓ denotes the sequence representation at layer ℓ , a block implements, at a coarse level, $X_\ell \mapsto X_{\ell+1}$.

We therefore ask:

Can learned state-transition computation be organized around explicit learned operations and state-conditioned applicability, with their controlled contributions composed into the transition as measurable and intervenable constituents of actual forward execution?

Autoregressive language modeling is the present constructive test domain, not a boundary on this research question.

At the broadest level, program execution and dynamical systems share the substrate $x \rightarrow x'$: the former treats computation as state or configuration change, while the latter treats the same abstract structure as state evolution. This commonality does not give DAWN-SRW a formal operational semantics. A more specific organizational motif, condition $\rightarrow$ operation $\rightarrow$ state transition, is likewise old; we do not claim it as novel. To separate its descriptive roles, we use the organizational coordinates

$$\mathcal{P} = (\mathcal{O}, C, \Gamma), \quad (1)$$

where $\mathcal{O} = \{O_i\}_i$ is an operation family, C is the state-conditioned applicability relation, and Γ is the composition rule. For scalar applicability,

$$C(x, O_i) = c_i(x), \quad F_{\mathcal{P}}(x) = \Gamma(\{c_i(x)O_i(x)\}_i). \quad (2)$$

These coordinates describe one family of explicit learned transition organizations, not a universal ontology or necessary decomposition. Here $\mathcal{O}$ specifies what transformations

can act; C specifies when and to what degree each is applicable; and Γ specifies how the controlled contributions $z_i(x) = c_i(x)O_i(x)$ form transition-producing computation. Thus operation identity, applicability, operation response, controlled contribution, and composition are distinct roles. Program/execution motivates conditioned operations, while dynamics/control provides mathematical precedent for continuously valued validity and composition of component effects. The *control–operation factorization* separates C from O ; Γ is a separate composition role, and normalized addition is only one DAWN-SRW realization of it.

The coordinates alone are not the scientific claim. An arbitrary mapping $T(x)$ is vacuously compatible with $O_1 = T$ and $c_1(x) = 1$. Such post-hoc decomposability is weaker than actual forward execution in which indexed operations, applicability variables, responses, controlled contributions, and specified composition semantics are architectural objects. This exposure, not the algebraic form alone, is the scientific target.

Within this broader problem, we call one program-oriented organizational formulation a *State-Conditioned Program*: the current state determines operation applicability, and controlled operation contributions are constituents of the actual transition-producing computation. This definition requires neither continuous or geometric control, sparse support, rank-1 operations, persistence, nor additive composition. The term *program* generalizes the analogy beyond symbolic source, discrete instructions, named human-readable primitives, call/return control flow, or formal operational semantics.

The same computation admits neural-map, dynamics, and State-Conditioned Program descriptions. The last asks which operations contribute, under what conditions and by how much, and what happens when contributions are changed. It is an actual-execution lens, not an ontological or privileged decomposition. Operational accounts of transitions between configurations provide its operation-oriented lineage (Plotkin, 1981), but no formal semantics for DAWN-SRW is claimed.

The resulting hierarchy is conceptual, not chronological. Its convergence point is explicitly organized learned state-transition computation, not the State-Conditioned Program formulation itself. Dynamics and control provide strong mathematical precedent for graded state-dependent validity, local transition models, and weighted composition (Zadeh, 1965; Takagi & Sugeno, 1985; Johansen & Foss, 1995; Rugh & Shamma, 2000). Learned conditional and modular systems then make both C_θ and $O_{\theta,i}$ learned computational objects (Shazeer et al., 2017; Rosenbaum et al., 2018; Goyal et al., 2021b; Rahaman et al., 2021); stable operation identity is an architectural property, whereas reuse is an empirical possibility. Read–write and fast-weight work supplies an

elementary transformation structure (Schlag et al., 2021; Lu et al., 2026a). The architectural hypothesis studied here is that learned state-transition computation can organize its actual forward execution as explicit, state-conditioned compositions of persistent learned operations. We use *state-conditioned operator composition* for the computational organization by which controlled operation contributions form a transition, not as a competing named formulation. We test that proposition constructively in autoregressive language modeling, not the novelty of any constituent ingredient.

Architectural exposure makes operation identities, applicability coefficients, responses, and controlled contributions direct measurement targets. Whether exposure supports useful causal manipulation is an empirical question, not an automatic consequence.

DAWN-SRW is one concrete geometric RW realization of this hypothesis. It turns state-conditioned control into a geometric applicability problem: a state-derived query defines a state-dependent neighborhood over persistent operator addresses, and continuously weighted RW effects within that neighborhood compose the transition. Applicability is controlled in a $d_c = 256$ space, while each RW operation reads the original D -dimensional representation and writes in $\mathbb{R}^D$; low-dimensional applicability control is therefore not a low-dimensional content read. Normalized addition composes Q/K/V and residual-state transition (RST) outputs. The implementation retains full candidate scope and dense RW execution. Geometry, rank-1 structure, persistence, candidate scope, and additive composition are realization choices, not formulation requirements.

The empirical study tests this realization through three ordered claims: viability, conditional organization, and causal accessibility. Fail-closed results delimit where the mechanistic localization procedure does not succeed; efficiency is not a headline claim.

Our contributions are:

1. **An organizational formulation for explicit learned state-transition computation and a constructive architectural test.** We connect operation-oriented program execution, continuous state-dependent dynamics, and learned modular computation as complementary precedents; use (O, C, Γ) as minimal organizational coordinates; formulate the State-Conditioned Program as one operation-oriented organization of these roles in actual forward execution; and test the resulting architectural hypothesis through DAWN-SRW in autoregressive language modeling. DAWN-SRW realizes continuous geometric applicability over persistent rank-1 RW operations whose controlled responses are explicitly composed into transition-producing computation.
2. **A constructive empirical test with bounded evi-**

dence. We test viability at 400M scale, report descriptive results for the 1.3B DAWN-SRW checkpoint (1.272B parameters), provide evidence of conditional organization through state-conditioned exact applicability support, and test causal accessibility through a discovery-frozen IOI circuit confirmed by suppression, matched controls, and exact restoration. Fail-closed results bound where the same localization procedure did not succeed.

2. Related Work

2.1. Operational views of computation and state transition

Structural operational semantics describes computation through transitions between configurations governed by rules (Plotkin, 1981). We use only the associated operation-oriented view: execution can be organized around identifiable operations and conditions under which they apply. DAWN-SRW does not define programming-language syntax, guarded commands, inference rules, loops, or formal operational semantics. Its overlap with dynamics begins at the abstract substrate $x \mapsto x'$, not at a claimed formal equivalence.

2.2. Continuous state-conditioned dynamics and local-model composition

Fuzzy membership, Takagi–Sugeno consequents, local model networks, and gain-scheduled or linear parameter-varying systems provide precedent for graded validity or scheduling and weighted component dynamics (Zadeh, 1965; Takagi & Sugeno, 1985; Johansen & Foss, 1995; Rugh & Shamma, 2000). Control-affine systems provide a complementary algebraic precedent: scalar controls multiply component vector fields and, under state feedback, become state-dependent coefficients (Sontag, 1998). DAWN-SRW instead uses learned internal applicability variables to organize forward-computation contributions; it is not a conventional control-affine system. Learned-dynamics work also exposes transition structure: dLDS learns overlapping dynamics components (Mudrik et al., 2024), Neural Hybrid Automata learn multi-mode dynamics and events (Poli et al., 2021), and concurrent LEON work realizes latent world-model transitions through context-modulated operator propagation and additive forcing (Lu et al., 2026b). These traditions supply mathematical precedent for state-dependent validity and weighted composition of component transition laws; the operation interpretation and actual-forward-execution exposure are the computational questions studied here. None of these ingredients is a novelty claim.

2.3. Learned conditional and modular computation

Input- or state-dependent routing and weighted learned transformations are established across MoE, convolutional, retrieval, and structured-routing systems (Shazeer et al., 2017; Fedus et al., 2022; Shen et al., 2023; Yang et al., 2019; He, 2024; Shi et al., 2026). Thus differentiable control over learned components is precedent; MoE-style conditional computation and continuous applicability are not novelty claims.

Stable or shared component identity across depth and context is likewise established in recurrent shared layers, global expert pools, routed attention, and compositional attention (Dehghani et al., 2019; Csordás et al., 2024a; Chen et al., 2026; Huang et al., 2026; Csordás et al., 2024b; Yang et al., 2025; Mittal et al., 2022). Such identity makes reuse measurable but does not by itself establish broad empirical reuse.

Routing and modular-mechanism work conditionally selects or composes learned functions, transition mechanisms, rules, and executors (Rosenbaum et al., 2018; Cases et al., 2019; Goyal et al., 2021b;a; Didolkar et al., 2021; Nguyen et al., 2023; Rahaman et al., 2021; Weiss et al., 2022). These precedents motivate a bridge from learned component-transition structure to persistent learned computational operations without identifying the two interpretations. Both C_θ and $O_{\theta,i}$ may be learned objects; persistence supplies identity, while cross-context reuse remains an empirical question.

2.4. Read–write and low-rank operation structure

Geva et al. interpret Transformer feed-forward layers as key-value memories with input-sensitive coefficients over output directions (Geva et al., 2021), and later decompose their additive updates into parameter-vector sub-updates traced through vocabulary-space predictions (Geva et al., 2022). Product Key Memory provides lineage for factorized keys and structured retrieval (Lample et al., 2019), scalable memory layers extend content-addressable trainable memories (Berges et al., 2024), Fast Weight Programmers provide an outer-product programming precedent (Schlag et al., 2021), and MoRAM uses rank-1 associative-memory experts with intrinsic keys (Lu et al., 2026a). Thus memory, retrieval, outer-product structure, read/write decomposition, and rank-1 experts are precedents. They provide the concrete transformation structure, not the general State-Conditioned Program definition. DAWN-SRW chooses $O_i(x) = \bar{w}_i(\bar{r}_i^\top x)$, then makes $c_i(x)O_i(x)$ an explicit controlled transition contribution under a separate applicability profile.

Implementation-level overlap in rank-1 experts, adaptive thresholds, geometric or low-dimensional control, read/write structure, or causal intervention does not imply shared conceptual ancestry, interpretation, or research program.

2.5. Neural programming as adjacent precedent

Neural Programmer-Interpreters, Neural Programmer, Fast Weight Programmers, and RASP provide adjacent precedents for neural programs, differentiable operation selection, instruction-linked computation, and program-level Transformer descriptions (Reed & de Freitas, 2016; Neelakantan et al., 2016; Schlag et al., 2021; Weiss et al., 2021). A State-Conditioned Program requires none of symbolic source, instruction traces, human-readable predicates, named primitive semantics, or formal operational semantics.

2.6. Causal intervention as evaluation methodology

IOI, AtP*, and RAVEL provide causal-intervention and localization precedent (Wang et al., 2023; Kramár et al., 2024; Huang et al., 2024). Here intervention validates a discovery-frozen RW circuit through suppression, controls, and exact restoration; it is evaluation methodology, not a constructive architectural ingredient or a claim of general semantic interpretability.

These traditions are neither historically linear nor mathematically identical: program/execution contributes operation-oriented organization; dynamics/control, state-dependent component effects and composition; learned modular computation, learned control and persistent components; and RW work, elementary read-write transformations. DAWN-SRW tests one intersection using continuous geometric applicability, persistent RW operations, and explicit composition; causal intervention supplies validation methodology.

3. DAWN-SRW

3.1. State Transitions and Control-Operation Factorization

Let

$$X_\ell = (x_{\ell,1}, \dots, x_{\ell,T}) \in \mathbb{R}^{T \times D}$$

denote the sequence representation at depth ℓ . A Transformer-like block produces state transitions through attention and a subsequent residual transformation:

$$Y_\ell = X_\ell + \text{Attn}_\ell(X_\ell), \quad (3)$$

$$X_{\ell+1} = Y_\ell + R_\ell(Y_\ell). \quad (4)$$

Equations 1–2 provide a coordinate-level organizational form, not a sufficient condition for computational organization or the architectural hypothesis itself. That hypothesis is that learned state-transition computation can organize its actual forward execution as explicit, state-conditioned compositions of persistent learned operations. In DAWN-SRW, this state-conditioned operator composition uses continuous geometric applicability over persistent RW operators and normalized additive composition. For a token state $x \in \mathbb{R}^D$

and computation type $a \in \{Q, K, V, \text{RST}\}$, define

$$F_a : \mathbb{R}^D \rightarrow \mathbb{R}^D. \quad (5)$$

Here F_Q, F_K, F_V parameterize relational interaction and F_{RST} parameterizes a direct residual-state update. Rank-1 RW operators, operation-address geometry, persistent identities, full candidate scope, dense physical execution, and normalized additive composition are realization choices, not requirements of either the formulation or the architectural hypothesis. We use *operation* for the abstract transformation, *operator* for an indexed learned architectural object implementing an operation, and *RW operator* for the rank-1 realization used here. Thus the computational formulation, architectural hypothesis, DAWN realization, and geometry implementation are distinct levels of claim.

3.2. Read-Write Operators

Each RW operator owns learned read and write parameters. With $\text{unit}(z) = z / \max(\|z\|_2, \epsilon)$, define normalized directions $\bar{r}_{a,i} = \text{unit}(r_{a,i})$ and $\bar{w}_{a,i} = \text{unit}(w_{a,i})$. The RW operator executes

$$O_{a,i}(x) = \langle x, \bar{r}_{a,i} \rangle \bar{w}_{a,i} = \bar{w}_{a,i} \bar{r}_{a,i}^\top x. \quad (6)$$

When nonzero, this is an elementary rank-1 read-write transformation, not a claim that rank-1 maps are universal causal atoms or the uniquely correct operation granularity. One controlled RW contribution can therefore be read as

$$c_{a,i}(x) O_{a,i}(x) = \underbrace{c_{a,i}(x)}_{\text{applicability}} \underbrace{(\bar{r}_{a,i}^\top x)}_{\text{scalar state read}} \times \underbrace{\bar{w}_{a,i}}_{\text{write direction}}. \quad (7)$$

Equation 7 reads as applicability $\times$ state read $\times$ write direction. Geometry controls when, where, and how much an operation contributes; it does not supply the operand. The read direction extracts an operation-specific scalar from the full representation, and the write direction determines its state change. Q/K/V RW operators compose attention interfaces; RST RW operators compose a direct residual-state update.

3.3. State-Conditioned Applicability

For each computation type, the current state induces an applicability profile over its N_a persistent RW operators,

$$C_a(x) = (c_{a,1}(x), \dots, c_{a,N_a}(x)), \quad (8)$$

whose coordinates determine whether and to what degree each RW operator participates. Section 3.5 defines the state-derived query $q_a(x)$ and operator address $k_{a,i}$. Their applicability score and applicability threshold are

$$s_{a,i}(x) = q_a(x)^\top k_{a,i}, \quad (9)$$

$$\tau_a(x) = -1 + 2\sigma(xu_a + \beta_a). \quad (10)$$

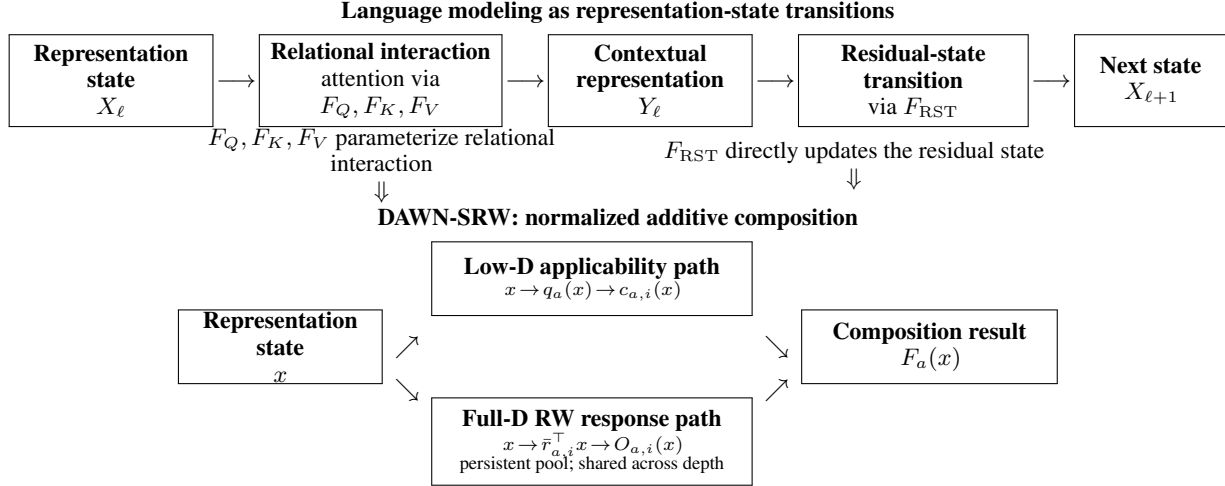


Figure 1. The two-level construction. Top: relational interaction and a residual-state transition update the representation state. Bottom: a low-D geometric condition produces applicability $c_{a,i}(x)$, while the full-D RW path reads $\bar{r}_{a,i}^\top x$ and writes along $\bar{w}_{a,i}$. Their product is a controlled operation contribution; normalized additive composition forms the transition-producing output $F_a(x)$. Operator pools persist across depth, candidate scope is full, and physical RW execution is dense.

The evaluated `linear_angular` rule defines

$$c_{a,i}(x) = \text{clip}\left(\frac{s_{a,i}(x) - \tau_a(x)}{\max(1 - \tau_a(x), 10^{-4})}, 0, 1\right). \quad (11)$$

We call $c_{a,i}(x) \in [0, 1]$ the *continuous-valued applicability* of operator i at state x . Its exact applicability support is

$$\begin{aligned} \mathcal{I}_a(x) &= \{i : c_{a,i}(x) > 0\} \\ &= \{i : s_{a,i}(x) > \tau_a(x)\}. \end{aligned} \quad (12)$$

When $s_{a,i}(x) \leq \tau_a(x)$, an operator contributes exactly zero; when $s_{a,i}(x) > \tau_a(x)$, $0 < c_{a,i}(x) \leq 1$ varies continuously with the applicability score. Thus the same applicability score and applicability threshold specify exact non-participation and continuous-valued applicability.

3.4. Normalized Additive Composition

DAWN-SRW uses normalized additive composition as one concrete realization of the composition rule Γ . The rule combines controlled RW-operator contributions; $F_a(x)$ denotes the resulting output, not the rule itself. With normalization factor $Z_a(x)$,

$$Z_a(x) = \left[\max\left(\sum_i c_{a,i}(x), 1\right) \right]^{p_a}, \quad (13)$$

$$F_a(x) = \frac{\sqrt{D/L}}{Z_a(x)} \sum_i c_{a,i}(x) \bar{w}_{a,i} \bar{r}_{a,i}^\top x. \quad (14)$$

Let $A_{a,i} = \bar{w}_{a,i} \bar{r}_{a,i}^\top$. Each term $c_{a,i}(x) A_{a,i} x$ is a controlled operation contribution. For $S_a(x) = \sum_i c_{a,i}(x) > 0$, define relative composition weights $\pi_{a,i}(x) = c_{a,i}(x)/S_a(x)$

and transition gain $g_a(x) = \sqrt{D/L} S_a(x)/Z_a(x)$. Then

$$\begin{aligned} A_{a,\text{eff}}(x) &= g_a(x) \sum_i \pi_{a,i}(x) A_{a,i}, \\ F_a(x) &= A_{a,\text{eff}}(x) x. \end{aligned}$$

Here $\pi_{a,i}(x) \geq 0$ and $\sum_i \pi_{a,i}(x) = 1$, separating relative operator composition from transition gain. If $S_a(x) = 0$, the effective operator and output are zero, and $\pi_{a,i}(x)$ need not be defined. This is an exact interpretation of the implementation, not a new architecture. At a fixed state, DAWN-SRW composes an effective transition operator from persistent RW operators; because $A_{a,\text{eff}}(x)$ depends on x , the map $x \mapsto F_a(x)$ is generally nonlinear, not globally linear. Normalized addition keeps controlled numerator terms separately measurable and intervenable. The experiments modify selected realized terms with $Z_a(x)$ fixed; Appendix A gives exact consequences.

3.5. Operation-Address Space

DAWN-SRW turns state-conditioned control into a geometric applicability problem: a state-derived query defines a state-dependent neighborhood over persistent operator addresses, and continuously weighted RW effects within that neighborhood compose the transition. This operation-address space is a controllable inductive bias, not part of the State-Conditioned Program definition. Representation and RW computation remain in $\mathbb{R}^D$, while $q_a(x), k_{a,i} \in \mathbb{R}^{d_c}$ are applicability-control variables. Once applicable, an operation still reads the original D -dimensional state through Equation 6. Thus similar applicability profiles can coexist with different full-D reads and transition contributions; this is an architectural possibility, not an established functional

equivalence. Here, $d_c = 256$ and

$$q_a(x) = \text{unit}(xP_a + b_a) \in \mathbb{R}^{d_c}. \quad (15)$$

A shared probe pair $P_r, P_w \in \mathbb{R}^{D \times d_c}$ maps functional read and write directions into the operator address

$$k_{a,i} = \text{unit}(\text{unit}(\bar{r}_{a,i}^\top P_r) \odot \text{unit}(\bar{w}_{a,i}^\top P_w)) \in \mathbb{R}^{d_c}. \quad (16)$$

Each operator address is a learned function of what its RW operator reads and where it writes, rather than an independently parameterized address. The operator address is not the scalar operand read by the RW operator: it is a low-dimensional coordinate used to evaluate applicability of the full-dimensional read–write transformation.

Because both $q_a(x)$ and $k_{a,i}$ are unit-normalized, they lie on the unit sphere $\mathbb{S}^{d_c-1}$. Writing $\theta_{a,i}(x)$ for the angle between them, $s_{a,i}(x) = q_a(x)^\top k_{a,i} = \cos(\theta_{a,i}(x))$, so $s_{a,i}(x) > \tau_a(x)$ is equivalent to $\theta_{a,i}(x) < \arccos(\tau_a(x))$. Thus the exact applicability support corresponds geometrically to a state-dependent spherical cap, or angular neighborhood, centered at $q_a(x)$ with angular radius $\arccos(\tau_a(x))$. Within this neighborhood, $c_{a,i}(x)$ gives continuous-valued applicability that varies with the cosine applicability score above the threshold and determines to what degree the operator contributes. This is the spherical realization of $(x, O_{a,i}) \mapsto c_{a,i}(x)$; exact implementation details are restated in Appendix A. The construction induces locality in address geometry, but the present measurements do not establish that geometric proximity corresponds to functional or causal similarity.

3.6. Attention-Mediated and Residual Transitions

For $\tilde{X}_\ell = \text{LN}_\ell^{(1)}(X_\ell)$, three normalized additive compositions produce the attention interfaces:

$$Q_\ell = F_Q(\tilde{X}_\ell),$$

$$K_\ell = F_K(\tilde{X}_\ell),$$

$$V_\ell = F_V(\tilde{X}_\ell),$$

$$A_\ell = \text{softmax}\left(\frac{Q_\ell K_\ell^\top}{\sqrt{d_h}} + M_{\text{causal}}\right),$$

$$\Delta X_{\text{attn},\ell} = A_\ell V_\ell W_\ell^O, \quad (17)$$

$$Y_\ell = X_\ell + \Delta X_{\text{attn},\ell}. \quad (18)$$

Q and K use a shared type-specific operator pool, while V uses a separate operator pool. A second normalization $\tilde{Y}_\ell = \text{LN}_\ell^{(2)}(Y_\ell)$ then queries the RST operator pool:

$$X_{\ell+1} = Y_\ell + F_{\text{RST}}(\tilde{Y}_\ell). \quad (19)$$

Thus RST applicability is evaluated from the contextualized post-attention state, and its normalized additive composition produces the direct residual-state update.

3.7. Persistent Operator Identity and Shared Control

Type-specific operator pools and their state-conditioned control parameterizations are shared across all L layers. Evolving normalized states therefore query the same operator pools rather than a separately parameterized profile at each depth. Persistence makes reuse measurable; it does not establish that broad cross-context reuse emerges. The two LayerNorms and attention output projection W_ℓ^O remain layer-specific; Appendix B gives the complete sharing inventory.

3.8. Candidate Scope and Physical Execution

For computation type a , the candidate scope is the full type-specific operator pool. Thus

$$\mathcal{I}_a(x) \subseteq \{1, \dots, N_a\},$$

with all N_a operators considered as candidates before applicability is determined. Applicability support is generally a strict subset, while physical RW execution is dense. Sharding, chunking, and accounting details appear in Appendix F.

4. Experimental Setup

The primary model is DAWN-SRW 400M (393.8M parameters, approximately 40B packed C4 tokens), compared with a parameter- and training-token-matched dense Transformer control (393.9M parameters). The 1.3B checkpoint, trained on approximately 20B tokens, is descriptive only. Complete architectural and training settings are in Appendix B.

Performance evaluation uses packed C4 validation and six zero-shot tasks: LAMBADA, HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and WinoGrande. The two 400M models use identical evaluation protocols; complete metrics, splits, and software details are in Appendix C.

For causal analysis, IOI discovery ranks operator sites and freezes a circuit before independent validation and held-out test. Suppression removes the frozen sites’ execution numerator, four matched-control families test specificity, and exact restoration reinserts the selected same-example numerator under the unchanged normalization factor. Appendix D gives the selection criteria, unit counts, statistics, and complete intervention protocol.

5. Results

5.1. Language-model performance

Table 1 consolidates the headline results. At 400M scale, DAWN-SRW differs from the matched dense control by +0.004318918 C4 cross-entropy and +0.003220495 six-task mean. These small measured gaps establish viability through performance retention, not superiority or formal

Table 1. Headline language-model results. The 1.3B row is descriptive.

Model	Parameters	C4 CE	Six-task mean
DAWN-SRW 400M	393.8M	2.940213	0.462872
Dense 400M	393.9M	2.935894	0.459652
DAWN-SRW 1.3B	1.272B	2.835879	0.485844

Table 2. Exact applicability support according to $c_{a,i}(x) > 0$. Q/K union counts an operator in the applicability support for Q or K at the same layer and token.

Computation	Pool	Mean support	Pool fraction
Q	4,600	487.896	10.6064%
K	4,600	386.722	8.4070%
Q/K union	4,600	812.388	17.6606%
V	14,500	1,649.994	11.3793%
RST	29,300	2,017.703	6.8864%

equivalence.

The descriptive 1.3B checkpoint reaches 2.835879 C4 cross-entropy and a 0.485844 six-task mean; without a matched dense control, this is not a scaling comparison. Appendix C reports C4 accuracy and perplexity, the complete task-level breakdown, and standard errors.

5.2. State-Conditioned Exact Applicability Support

At all 9,994,240 packed-C4 positions, Q, K, V, and RST each have strict-subset applicability support according to $c_{a,i}(x) > 0$ (Table 2). This provides evidence of conditional organization through state-conditioned exact applicability support under Equation 11, with strict-subset rather than full-pool participation.

Restricting RW execution to exact applicability support would substantially reduce static accounting, but the resulting estimate remains above the matched dense control; this is not measured latency, energy, or hardware speedup. Detailed accounting and Figure 2 are in Appendix F.

5.3. Frozen IOI circuit and causal intervention

Discovery and confirmation are separated: a 4,096-site Q/K-centered circuit is frozen before validation, and held-out test reuses the unchanged specification. Complete discovery statistics are in Appendix D.

Suppression causes large behavior changes on both splits and exceeds every replicate in all four matched-control families. Activation-matched controls are strongest; frozen-minus-control margin separation is 2.464297 on validation and 1.937754 on test (both $p = 0.00049975$). Exact restoration recovers more than 99.8% of the suppression effect. Together these interventions establish causal accessibility for the evaluated IOI behavior; they do not imply a unique

or human-readable algorithm.

5.4. Fail-Closed Results

Fail-closed results on RAVEL, ARC, and 1.3B IOI arise from distinct preregistered criteria: localization instability, failed causal suppression, and insufficient cumulative importance, respectively. Exact statistics and decisions are in Appendix E and Table 9.

6. Discussion

6.1. What the results establish

The narrow DAWN-SRW hypothesis is that useful learned transitions can arise by continuously composing persistent RW operations under learned geometric applicability. The experiments support only the three bounded claims: matched-400M viability, conditional organization, and causal accessibility in a discovery-frozen IOI circuit. Fail-closed studies delimit unsuccessful localization; causal access implies neither semantics nor reusable programs.

Architectural exposure. The present construction highlights five architectural properties that support direct measurement and localized intervention: referable indexed operations O_i in actual forward execution; explicit applicability $c_i(x)$; separately accessible controlled contributions $c_i(x)O_i(x)$; specified composition semantics Γ ; and intervention semantics stating what changes and what is held fixed. These are properties of the tested exposure, not universal axioms or necessary conditions.

Exposure and intervention. Exposure does not guarantee useful causal manipulability. The experiments use a *contribution intervention*: selected realized numerator terms are suppressed or restored with $Z_a(x)$ fixed. A *control intervention* would alter applicability or control and recompute downstream composition; it is not tested. Control-profile transplant and recomposed counterfactual execution remain open.

6.2. From the Computational Question to an Open Design Space

Making learned state-transition organization explicit exposes four design axes: control and applicability, operation and identity, coupling and composition, and candidate scope and physical execution. DAWN-SRW supplies one tested point, not requirements for this broader design space.

Control and applicability. Does control geometry encode transition requirements or merely useful applicability? The $d_c = 256$ sphere is one option, not an established functional or causal geometry.

Table 3. Frozen IOI circuit results. Accuracy arrows show intact to suppressed. Recovery is the fraction of the suppression effect recovered by exact same-example numerator restoration.

Split	Units	Intact margin	Suppressed margin	Drop (95% CI)	Accuracy	Recovery / restored acc.
Validation	123	3.465456	-4.399331	7.864787 [7.490080, 8.241263]	1.000000 → 0.008130	0.999661 / 0.991870
Held-out test	114	3.254720	-4.155613	7.410332 [7.016485, 7.818472]	1.000000 → 0.008772	0.998406 / 1.000000

Operation and identity. What granularity and persistence should operations have? Identity makes reuse measurable but establishes neither broad reuse, a single function, nor semantic purity.

Coupling and composition. How much interaction can composition add while preserving operation-wise measurement and intervention? Normalized addition offers high separability; structured nonlinear interaction trades some attribution for interaction; opaque mixing can retain identities while obscuring individual contributions. Sequential or hierarchical composition may also have noncommutative semantics.

Candidate scope and physical execution. How narrowly can candidate scope and execution be restricted without losing quality or organization? Exact applicability support does not imply sparse execution: the evaluated model considers the full operator pool and executes RW densely. Hardware-exploitable sparse retrieval or execution remains open.

6.3. Structural Questions Exposed by the Organization

The evaluated model uses $d_c = 256$ control while RW responses act in full D . How small can control geometry be? Does spherical applicability recover transition regimes? Do persistent identities become specialist, reusable, or recombinable? Does similar organization predict similar causal effects? These remain open; neither minimal control dimension nor geometry–function or geometry–causality correspondence is established.

6.4. Limitations

Global candidate evaluation and dense RW execution remain expensive; even static accounting restricted to exact applicability support is 2.1168 times the matched dense control’s backbone. No production latency, energy, or hardware advantage is measured.

The state–condition–operation motif is not novel, and post-hoc functional decomposition is insufficient. Neither the State-Conditioned Program, its control–operation factorization, spherical geometry, $d_c = 256$, nor geometry–function correspondence is claimed to be intrinsic, unique, necessary, privileged, or optimal. The low-dimensional applicability path is not shown to be minimal, sufficient, a statistical quotient, or evidence that local execution is generally

low-dimensional; it may control broadly distributed full-dimensional operations. Persistence makes reuse measurable but does not establish it.

Causal evidence is confined to the specified IOI behavior. Fail-closed RAVEL, ARC, and 1.3B IOI results delimit mechanistic generalization; the differently budgeted 400M and 1.3B checkpoints are not a controlled scaling curve; and operators need not map one-to-one to human-readable concepts. Suppression/restoration establishes neither arbitrary program editing, semantic interpretability, operator replacement, nor control transplant.

7. Conclusion

Learned state-transition computation is usually organizationally implicit. The broader problem is explicit learned state-transition organization; the coordinates $\mathcal{P} = (\mathcal{O}, C, \Gamma)$ describe one minimal family, and a State-Conditioned Program is one program-oriented actual-execution formulation. The architectural hypothesis is that learned forward computation can organize as state-conditioned operator composition over persistent learned operations. DAWN-SRW tests one geometric rank-1 RW realization in autoregressive language modeling: a state-derived $d_c = 256$ spherical neighborhood supplies continuous applicability, persistent RW operations retain full- D read–write access, and normalized addition explicitly composes the transition-producing output.

The evidence is bounded: measured 400M viability, descriptive 1.3B results, evidence of conditional organization from state-conditioned exact applicability support, and causal accessibility through a discovery-frozen IOI circuit tested by suppression, matched controls, and exact restoration. Fail-closed results mark where localization did not succeed.

The main interpretation is not that the decomposition or geometry is intrinsic, unique, semantically pure, or computationally optimal. DAWN-SRW provides one constructive instance in which a high-dimensional state defines a low-dimensional geometric condition over persistent operations, while applicable operations retain full-dimensional read–write access and compose the transition. The broader question is how learned computation can preserve continuous neural flexibility while exposing referable operations, applicability, controlled contributions, and composition semantics for measurement and intervention.

A. Exact Operation-Address and Composition Equations

This appendix restates the production computation as a compact reference. Raw read and write parameters are normalized in float32 before being cast for dense contractions:

$$\bar{r}_i = \frac{r_i}{\max(\|r_i\|_2, \epsilon)}, \quad \bar{w}_i = \frac{w_i}{\max(\|w_i\|_2, \epsilon)}.$$

One pair of model-level probes is shared across QK, V, and RST. In canonical notation, the operator address is

$$k_{a,i} = \text{unit}(\text{unit}(\bar{r}_{a,i}^\top P_r) \odot \text{unit}(\bar{w}_{a,i}^\top P_w)).$$

The query and applicability threshold use separate direct projections. The `DirectTau` projection produces the implementation variable `raw_tau` = $xu_a + \beta_a$, whose transformed value is the canonical applicability threshold $\tau_a(x)$:

$$q_a(x) = \text{unit}(xP_a + b_a), \\ \tau_a(x) = -1 + 2\sigma(xu_a + \beta_a), \quad s_{a,i}(x) = q_a(x)^\top k_{a,i}.$$

For the `linear_angular` applicability rule:

$$c_{a,i}(x) = \text{clip}\left(\frac{s_{a,i}(x) - \tau_a(x)}{\max(1 - \tau_a(x), 10^{-4})}, 0, 1\right)$$

and padding rows are masked to zero. This is the continuous-valued applicability used in normalized additive composition. With normalization factor $Z_a(x) = \max(\sum_i c_{a,i}(x), 1)^{p_a}$, the production composed output before causal attention or residual addition is

$$F_a(x) = \sqrt{\frac{D}{L}} \frac{\sum_i c_{a,i}(x) \langle x, \bar{r}_{a,i} \rangle \bar{w}_{a,i}}{Z_a(x)},$$

with $p_Q = p_K = 0.5$, $p_V = 1.0$, and $p_{\text{RST}} = 1.2$. Operator rows are sharded over the model axis; local numerators and local sums of applicability values are reduced across shards to produce the global numerator and normalization factor. Chunking changes the contraction schedule, not the equation.

Relative composition, convex hull, and pointwise norm.

For $S_a(x) = \sum_i c_{a,i}(x) > 0$, let $\pi_{a,i}(x) = c_{a,i}(x)/S_a(x)$, $A_{a,i} = \bar{w}_{a,i} \bar{r}_{a,i}^\top$, and $\tilde{A}_a(x) = \sum_i \pi_{a,i}(x) A_{a,i}$. Then

$$\tilde{A}_a(x) \in \text{conv}\{A_{a,i} : i \in \mathcal{I}_a(x)\},$$

$$A_{a,\text{eff}}(x) = g_a(x) \tilde{A}_a(x).$$

Thus applicability support selects participating operators, relative weights choose their convex mixture in matrix space, and $g_a(x)$ supplies transition gain. Because $\text{unit}(z) = z / \max(\|z\|_2, \epsilon)$ has norm at most one, $\|A_{a,i}\|_2 \leq 1$, hence $\|\tilde{A}_a(x)\|_2 \leq 1$ and $\|A_{a,\text{eff}}(x)\|_2 \leq g_a(x)$. These are pointwise² fixed-state facts, not a global Lipschitz bound on $x \mapsto F_a(x)$; they also imply no correspondence between address-space and matrix-space proximity. When $S_a(x) = 0$, $A_{a,\text{eff}}(x) = 0$ and the relative weights are undefined.

Local Jacobian and update-span consequence. Fix x and abbreviate $\alpha = \sqrt{D/L}$, $c_i = c_{a,i}(x)$, $u_i = \bar{r}_{a,i}^\top x$, $Z = Z_a(x)$, and $g_i = c_i u_i / Z$. On a neighborhood with fixed exact applicability support, and at points away from clipping, support, and normalization kinks, the realized normalized RW composition has

$$\nabla_x g_i = Z^{-1}(c_i \bar{r}_{a,i} + u_i \nabla_x c_i) - Z^{-2} c_i u_i \nabla_x Z, \\ J_{F_a}(x) = \alpha \sum_{i \in \mathcal{I}_a(x)} \bar{w}_{a,i} \nabla_x g_i^\top.$$

For $\mathcal{W}_a(x) = \text{span}(\{\bar{w}_{a,i}\}_{i \in \mathcal{I}_a(x)})$, consequently

$$\text{Im } J_{F_a}(x) \subseteq \mathcal{W}_a(x), \quad \text{rank } J_{F_a}(x) \leq \dim \mathcal{W}_a(x).$$

For a residual transition $T(x) = x + F_a(x)$, the same containment applies to $J_T(x) - I$, not generally to $J_T(x)$. This is a local consequence of the evaluated normalized additive RW realization under the stated regularity conditions, not a theorem about every State-Conditioned Program.

B. Complete Model and Training Settings

The 400M DAWN-SRW checkpoint is the primary performance and mechanistic model. The dense 400M checkpoint is only a matched performance control. The 1.3B DAWN-SRW checkpoint is used for descriptive performance and one IOI localization replication that produced a fail-closed result. The still-incomplete dense 1.3B run is excluded.

The QK, V, and RST operator pools; the two RW-to-address probes; the Q/K/V query projection; the RST query projection; and their `DirectTau` projections are shared across all layers. Q and K share read/write rows but retain separate query slices, applicability thresholds, applicability values, and outputs. The two LayerNorms and attention output projection W_ℓ^O are layer-specific; positional and token embeddings and the final normalization are model-level parameters. Depth is not an explicit operator-pool index: it changes the representation state presented to the shared control interface.

C. Benchmark Metrics and Protocol Details

C4. All reported C4 rows use the same packed validation artifact and `packed_c4_validation_v1` protocol. The capped view contains 19,531 sequences. Exactly 19,520 sequences form 610 complete global evaluation batches, producing 9,974,720 valid next-token targets; 11 incomplete-tail sequences are excluded before scoring.

Table 4. Model and training specification. The 1.3B row is descriptive and does not form a controlled scaling curve with the 400M rows.

Model	Parameters	Tokens	D	Layers	Heads	QK / V / RST rows	Checkpoint
DAWN-SRW 400M	393,800,708	39,999,504,384	2,304	18	36	4,600 / 14,500 / 29,300	76,293
Dense Transformer 400M	393,896,960	39,999,504,384	1,280	18	20	—	76,293
DAWN-SRW 1.3B	1,272,038,404	19,999,981,568	4,096	24	32	8,534 / 26,904 / 54,352	87,193

Table 5. Detailed settings from the registered training configurations and restored checkpoint manifests.

Setting	DAWN-SRW 400M	Dense 400M	DAWN-SRW 1.3B
Tokenizer vocabulary	30,522	30,522	30,522
Residual width	2,304	1,280	4,096
Layers	18	18	24
Heads / head width	36 / 64	20 / 64	32 / 128
Maximum sequence length	512	512	512
Feed-forward width	—	5,120	—
Control dimension d_c	256	—	256
QK / V / RST operator-pool size	4,600 / 14,500 / 29,300	—	8,534 / 26,904 / 54,352
QK / V / RST chunks	1 / 1 / 4	—	1 / 2 / 8
QK / V / RST normalizer exponent p_a	0.5 / 1.0 / 1.2	—	0.5 / 1.0 / 1.2
Initial applicability-support fraction targets	0.05 / 0.03 / 0.01	—	0.05 / 0.03 / 0.01
Training batch	1,024	1,024	448
Mesh (data $\times$ model)	16×2	16×2	16×2
Learning rate	3×10^{-4}	3×10^{-4}	2×10^{-4}
Warmup ratio	0.08	0.08	0.06
Weight decay	0.1	0.1	0.1
Applicability-threshold learning-rate multiplier	0.001	—	0.001
Dropout / <code>router.dropout</code>	0 / 0	0 / —	0 / 0
Gradient checkpointing	yes	yes	yes
Seed	1	1	1

Table 6. Packed C4 validation on the identical 9,974,720 target tokens.

Model	CE	PPL	Accuracy
DAWN-SRW 400M	2.940213	18.919873	0.424912
Dense 400M	2.935894	18.838336	0.427100
DAWN-SRW 1.3B	2.835879	17.045382	0.437099

Zero-shot. The protocol is `mamba_table3_zero_shot_v1`, using `stock_lm-eval==0.4.2`, global batch 32, maximum length 512, length buckets 64/128/256/512, and zero few-shot examples. Evaluated split sizes are 5,153 for LAMBADA, 10,042 for HellaSwag, 1,838 for PIQA, 2,376 for ARC-Easy, 1,172 for ARC-Challenge, and 1,267 for WinoGrande. Task, dataset, tokenizer, and software settings are identical between the two 400M models.

C.1. Detailed benchmark results

LAMBADA perplexities are 24.0774 ± 0.8005 for DAWN-SRW 400M, 24.0042 ± 0.8073 for dense 400M, and 17.6358 ± 0.5390 for DAWN-SRW 1.3B.

D. IOI Localization and Intervention Details

D.1. Frozen discovery

Discovery admits 119 paired-correct independent units and ranks 456,697 sites. Split-half rank stability is 0.913978765. The frozen selection rule is the smallest prefix satisfying

cumulative absolute importance at least 0.70 and split top- k overlap at least 0.95. The selected $k = 4096$ prefix has:

- cumulative absolute importance 0.719248713;
- 3,993 shared split sites, overlap 0.974853516, and Jaccard 0.950940700;
- $Q/K/V/RST$ counts = 1744/2350/2/0;
- coverage of every layer from 0 through 17.

Validation and test do not recompute or alter this selection.

D.2. Suppression, controls, and restoration

Let T_{corr} and T_{src} denote the correct-answer and source-answer token sets. The primary score is

$$\mathcal{M} = \sum_{t \in T_{\text{corr}}} \log p(t) - \sum_{t \in T_{\text{src}}} \log p(t).$$

Suppression zeros the frozen sites’ execution numerator at specified token positions. It leaves applicability and the normalization factor unchanged. Controls use the same 4,096-site count and exclude all circuit sites. Activation matching conditions on layer, computation type, and the discovery mean pre-scale operator-output-norm quantile. Restoration executes a distinct intervention: after suppression, it adds back the exact selected same-example numerator under the unchanged normalization factor.

Table 7. Zero-shot accuracy. Parentheses contain standard errors. HS, AE, and AC denote HellaSwag, ARC-Easy, and ARC-Challenge.

Model	LAMBADA	HS	PIQA	AE	AC	WinoGrande	Mean
DAWN-SRW 400M	0.352028 (0.006654)	0.458375 (0.004972)	0.712187 (0.010563)	0.484007 (0.010255)	0.256826 (0.012767)	0.513812 (0.014047)	0.462872
Dense 400M	0.353192 (0.006659)	0.457080 (0.004971)	0.698041 (0.010712)	0.475589 (0.010248)	0.249147 (0.012639)	0.524862 (0.014035)	0.459652
DAWN-SRW 1.3B	0.395886 (0.006813)	0.510157 (0.004989)	0.723069 (0.010440)	0.509680 (0.010258)	0.268771 (0.012955)	0.507498 (0.014051)	0.485844

Table 8. Control effects and frozen-minus-control separation. Each control family contains 100 replicates. All paired permutation p -values are 0.0004997501 with 2,000 permutations.

Control	Validation drop	Frozen minus control (95% CI)	Test drop	Frozen minus control (95% CI)
Equal-count	0.070886	7.793902 [7.437851, 8.161572]	0.080276	7.330057 [6.922408, 7.716879]
Same-layer	0.039389	7.825399 [7.440629, 8.180288]	0.041073	7.369260 [6.977447, 7.754539]
Activation-matched	5.400490	2.464297 [2.076200, 2.832697]	5.472579	1.937754 [1.564818, 2.319706]
Route-frequency	0.219162	7.645626 [7.289512, 8.031625]	0.242665	7.167668 [6.779836, 7.545537]

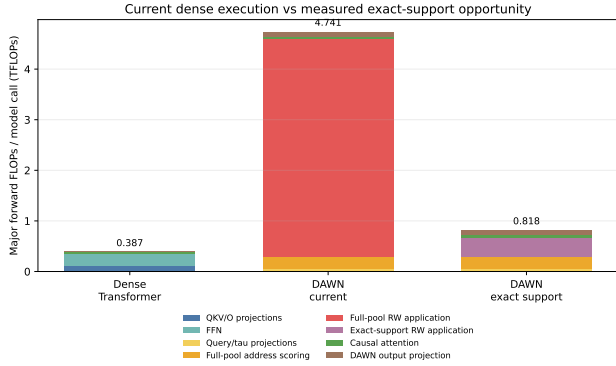


Figure 2. Static backbone accounting at batch 1 and sequence length 512. Address scoring remains dense in the estimate restricted to exact applicability support. Current dense RW execution also applies all full-dimensional RW rows. No production latency claim is made.

On validation, restoration returns the margin to 3.462791, leaving residual drop 0.002665. On held-out test, it returns the margin to 3.242905, leaving residual drop 0.011815. The respective recovery fractions are 0.999661155 (95% CI [0.998654544, 1.000584125]) and 0.998405645 (95% CI [0.997323737, 0.999385266]).

E. Fail-Closed Results

RAVEL. Discovery uses 142 independent units. Rank stability is 0.628762 for Continent, 0.676243 for Country, and 0.660622 for Language, below the preregistered requirement. No prefix passes the complete discovery rule, no variable circuit is frozen, and validation, transfer, and test are not opened.

ARC. Discovery stability is 0.89386 and a 4,096-site Q/K-only circuit is frozen. On 51 validation units, intact margin is 1.735534 and suppressed margin is 2.819139, giving a frozen margin drop of -1.083605 (95% CI $[-1.472180, -0.690348]$). The effect has the wrong sign and fails control separation. Restoration recovery is 1.004083 (95% CI

[0.994587, 1.013652]), but restoration alone does not pass the conjunction. Held-out test remains closed.

1.3B IOI. The scale replication admits 125 independent units and passes rank stability at 0.907187749. The largest allowed 4,096-site prefix reaches only 0.635144 cumulative absolute importance, below the 0.70 criterion. No circuit is frozen; validation and test remain closed.

F. Compute Accounting

The implementation shards operator rows across the model mesh and processes them in dense chunks. The 400M configuration uses one QK chunk, one V chunk, and four RST chunks. Every query scores all local operator addresses, and every operator row participates in dense read and write contractions before cross-shard reduction. RW operators execute in the full residual dimension D ; the d_c -dimensional operation-address space is used only for applicability.

The applicability-support analysis evaluates 9,994,240 packed-C4 positions and observes applicability scores and applicability thresholds before RW application. It stores aggregate statistics, not per-token operator identities or vectors. Static accounting uses batch 1, sequence length 512, and 1 MAC = 2 FLOPs. It counts major matmul/einsum operations and omits softmax, normalization, applicability elementwise arithmetic, embedding lookup, residual additions, and communication collectives.

Static forward accounting is 4.741260116 TFLOPs for current dense DAWN-SRW, 0.818230250 TFLOPs for dense addressing plus RW application restricted to exact applicability support, and 0.386547057 TFLOPs for the matched dense control. The estimate restricted to exact applicability support is 17.2577% of current DAWN-SRW accounting but still 2.1168 times the dense Transformer. These are static arithmetic estimates, not measured latency, energy, or hardware speedup.

The estimate restricted to exact applicability support is mathematically output-equivalent if the executor can skip all and

Table 9. Fail-closed results. Later phases were not opened after a failed preregistered criterion.

Study	Passed	Failed criterion	Decision
RAVEL 400M	Behavioral eligibility	Variable stability: 0.628762 / 0.676243 / 0.660622	No frozen circuit; no validation, transfer, or test
ARC 400M	Stable discovery; 4,096 Q/K sites; restoration	Suppression effect wrong sign; control separation failed	Reject causal replication; held-out test closed
IOI 1.3B	Behavioral eligibility; rank stability 0.907188	4,096-site cumulative importance 0.635144 < 0.70	No frozen circuit; validation and test closed

Table 10. Backbone static accounting. “Exact applicability support” retains full dense address scoring and replaces only full-pool RW application with the measured applicability support according to $c_{a,i}(x) > 0$.

Component	Current dense DAWN-SRW	Dense address + exact RW	Dense Transformer
Address scoring + query/threshold projections	0.293742	0.293742	–
RW application	4.306187	0.383157	–
Causal attention	0.043487	0.043487	0.024159
Dense output / QKV / FFN maps	0.097845	0.097845	0.362388
Total TFLOPs	4.741260	0.818230	0.386547

only zero-weight RW terms. It does not include indexed address retrieval and does not demonstrate kernel parity. Instrumented profiling time is excluded from latency claims because histogram collection is part of the measured path.

G. Reproducibility Identities

The experiments reported here use DAWN-SRW implementation revision v4.1.7.2, recorded under the repository model identity `spatial-rl-v4.1.7.2`; literal repository filenames and artifact paths retain the historical shorthand v4172.

The machine-readable checkpoint registry is `configs/paper_checkpoint_registry.yaml`. Exact evaluation commits, software versions, data hashes, result roots, and complete artifact hashes are recorded in the repository experiment ledger. Historical experiment branches are evidence identities only; this source is maintained on the canonical main branch.

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Table 11. Primary immutable identities. Branch names are omitted because they move; exact checkpoint steps, hashes, and artifact locations are the binding identities.

Evidence	Exact identity	Repository source
DAWN-SRW 400M	model spatial-r1-v4.1.7.2; step 76,293; checkpoint identity a7ce8afc...f91c17; config hash 08733ae4...5039b	configs/train_config_v4172_400M_c4_40B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
Dense 400M	model baseline-JAX; step 76,293; config hash c3ce5d72...82fc25	configs/train_config_baseline_tpu_400M_c4_40B_v4_64_tp2.yaml
DAWN-SRW 1.3B	model spatial-r1-v4.1.7.2; step 87,193; checkpoint identity 2e38b41f...bf5f9; config hash 1e877263...4a5222	configs/train_config_v4172_1p3B_c4_20B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
IOI circuit	parsed frozen-spec hash 6ec26618...418c; site hash 573cfff8...0200; selected-row hash 1439f6f3...fe0c1	configs/paper_ioi_frozen_circuit_v4172_400m.yaml
Compute support	support hash b1eac5f5...bdc2; accounting hash d7ad6115...984e	assets/paper/v4172_400m_compute_support_step76293/

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